

FCC Fact Sheet*
Modernizing Spectrum Sharing for Satellite Broadband
Notice of Proposed Rulemaking – SB Docket No. 25-157

Background: Recent years have witnessed unprecedented growth and innovation in the satellite marketplace, driven by an explosion in the deployment of non-geostationary satellite systems. These systems have already begun to offer advanced services, including high-speed, low-latency broadband connectivity across the country to American consumers and businesses. These systems rely on shared spectrum and must operate compatibly with other satellite systems and with terrestrial networks. In the 10.7-12.7 GHz, 17.3-18.6 GHz, and 19.7-20.2 GHz bands, non-geostationary satellite systems are required to meet equivalent power-flux density limits developed internationally in the 1990s for the protection of geostationary networks. These limits directly affect the efficiency of spectrum use by broadband satellite constellations today and thus are limiting the level of satellite broadband service available in the marketplace.

What the Notice of Proposed Rulemaking Would Do:

- Initiate a review of the decades-old spectrum sharing regime between geostationary and non-geostationary satellite systems operating in the 10.7-12.7, 17.3-18.6, and 19.7-20.2 GHz bands.
- Invite comment on how satellite technology and operations have changed since equivalent power-flux density limits were developed and adopted, how the current limits protect geostationary networks and impact the services provided by non-geostationary systems, and whether alternative sharing frameworks would promote greater efficiency to the benefit of American consumers.
- Invite comment on what short-term and long-term protection criteria would be appropriate for today's geostationary networks, and which reference links should be used to assess compliance with the protection criteria.
- Invite comment on the potential for improved information sharing among geostationary and non-geostationary satellite operators to enhance efficient spectrum sharing, how to address concerns related to aggregate interference, and how to ensure continued protection of co-frequency terrestrial stations.
- Grant a petition for rulemaking filed by Space Exploration Holdings, LLC, to initiate this proceeding and deny the oppositions to the petition.

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**Before the
Federal Communications Commission
Washington, D.C. 20554**

In the Matter of	)	
	)	
Modernizing Spectrum Sharing for Satellite	)	SB Docket No. 25-157
Broadband	)	
	)	RM-11990
Revision of the Commission's Rules to Establish	)	(Terminated)
More Efficient Spectrum Sharing between NGSO	)	
and GSO Satellite Systems	)	

NOTICE OF PROPOSED RULEMAKING*

Comment Date: [[45 days after publication in the Federal Register]]

Reply Comment Date: [[75 days after publication in the Federal Register]]

Adopted: []

Released: []

By the Commission:

I. INTRODUCTION

1. The United States' pursuit of space leadership demands that the Commission's rules stay ahead of the rapidly unfolding space innovations that are providing massive consumer benefit to Americans. Our nation's commercial space industry is already delivering services that are changing lives, from connecting rural families with high-speed, low latency broadband to enabling life-saving communications in rural places as well as forests, mountains, and wilderness areas. And every indication is that this is only at the beginning. In a matter of years – with the right framework in place – the U.S. space industry will be delivering even faster high-speed services, will bring more competition and choice for consumers, and will support entirely new categories of innovation to keep America the strongest and safest nation on Earth. It is imperative the Commission does everything possible to clear the way for American innovation and investment in space excellence.

2. With this goal at the forefront, in this Notice of Proposed Rulemaking, we initiate a review of the decades-old spectrum sharing regime between geostationary (GSO) and non-geostationary (NGSO) satellite systems operating in the 10.7-12.7, 17.3-18.6, and 19.7-20.2 GHz bands that represents the single most constraining regulatory requirement on NGSO satellite systems currently deploying at breakneck speed.¹ By taking a fresh look at today's satellite technology and operations, this proceeding

* This document has been circulated for tentative consideration by the Commission at its April 28, 2025 open meeting. The issues referenced in this document and the Commission's ultimate resolution of those issues remain under consideration and subject to change. This document does not constitute any official action by the Commission. However, the Chairman has determined that, in the interest of promoting the public's ability to understand the nature and scope of issues under consideration, the public interest would be served by making this document publicly available. The Commission's *ex parte* rules apply and presentations are subject to "permit-but-disclose" *ex parte* rules. See, e.g., 47 CFR §§ 1.1206, 1.1200(a). Participants in this proceeding should familiarize themselves with the Commission's *ex parte* rules, including the general prohibition on presentations (written and oral) on matters listed on the Sunshine Agenda, which is typically released a week prior to the Commission's meeting. See 47 CFR §§ 1.1200(a), 1.1203.

(continued....)

will ensure highly efficient and effective use of the shared spectrum, and support a more efficient and competitive market for satellite broadband and other in-demand services while uncapping the potential of satellite constellations that were unthinkable when the current regime was developed, to the ultimate benefit of American consumers.

II. BACKGROUND

3. *Overview.* Recent years have witnessed unprecedented growth and innovation in the satellite marketplace. New NGSO satellite operators have launched thousands of satellites in the span of a few years,² and have begun to offer high-speed, low-latency broadband services.³ Established GSO satellite operators continue to deploy powerful, new satellites with enhanced capabilities.⁴ By one estimate, the supply of high-throughput satellite capacity tripled between 2021 and 2023, with NGSO operators accounting for over 90% of the net supply.⁵ GSO and NGSO satellites are helping bridge the digital divide.⁶

4. These services rely on a spectrum access system predicated on different systems – satellite as well as terrestrial – sharing the same spectrum bands and rules that enable them to coexist. In frequency bands between 10.7 GHz and 30 GHz, NGSO fixed-satellite service (FSS) systems share FSS allocations with GSO FSS networks, and must also operate compatibly with broadcasting-satellite service (BSS) networks and stations in other services, including terrestrial services.⁷ NGSO FSS systems must comply with power limits expressed in equivalent power-flux density (EPFD)⁸ to protect GSO FSS and BSS networks,⁹ and separate power limits expressed in power-flux density (PFD) to protect terrestrial services.¹⁰ Within the 10.7-30 GHz range, NGSO FSS EPFD downlink limits apply in the 10.7-12.7

¹ As discussed below, we grant the petition to initiate this rulemaking and deny the oppositions to the petition. See SpaceX Petition for Rulemaking, Revision of the Commission's Rules to Establish More Efficient Spectrum Sharing between NGSO and GSO Satellite Systems, RM-11990 (filed Aug. 9, 2024) (Petition); Viasat Opposition; EchoStar Opposition.

² See *Communications Marketplace Report*, GN Docket No. 24-119, FCC 24-136, paras. 184, 194, 425 (rel. Dec. 31, 2024) (2024 *Marketplace Report*).

³ See *id.*, paras. 179, 184.

⁴ See *id.*, paras. 184, 194, 425.

⁵ Press Release, Euroconsult, Non-Geostationary Orbit Constellations Redefining the High Throughput Satellites Market Landscape (Apr. 25, 2024), <https://www.euroconsult-ec.com/press-release/non-geostationary-orbit-constellations-redefining-the-high-throughput-satellites-market-landscape/>.

⁶ 2024 *Marketplace Report*, FCC 24-136, para. 185. In 2023, for example, GSO satellite operator Hughes and NGSO satellite operator SpaceX each served over one million subscribers in the United States. *Id.* NGSO satellite licensee Kuiper has announced it will begin full-scale deployment of its satellite internet network of 3,232 satellites with a launch of 27 satellites in April 2025. See <https://www.aboutamazon.com/news/innovation-at-amazon/project-kuiper-satellite-internet-first-launch>.

⁷ See 47 CFR § 2.106. The bands between 17.7 GHz and 20.2 GHz are also subject to coordination with Federal Government users. 47 CFR § 2.106(a), (c)(334).

⁸ EPFD is the sum of the power flux densities produced at a GSO receive station on the Earth's surface or in the geostationary orbit, as appropriate, by all the transmit stations within an NGSO FSS system, taking into account the off-axis discrimination of a reference receiving antenna assumed to be pointing in its nominal direction. 47 CFR § 25.103.

⁹ 47 CFR § 25.146(a)(2), (c); see also 47 CFR § 25.289. In the 18.8-19.3 GHz and 28.6-29.1 GHz bands, EPFD limits do not apply and GSO FSS networks must not cause harmful interference to, or claim protection from, NGSO FSS systems. 47 CFR § 2.106(a), (d)(165).

¹⁰ 47 CFR § 25.146(a)(1).

GHz, 17.3-18.6 GHz, and 19.7-20.2 GHz bands in the United States.¹¹ Applicants for NGSO FSS space station licenses, and non-U.S.-licensed satellite operators seeking access to the U.S. market, must certify that they will comply with the specified EPFD limits.¹² Prior to initiating service, NGSO FSS operators' EPFD showings submitted to the International Telecommunication Union (ITU) must receive a "favorable" or "qualified favorable" finding by the ITU Radiocommunication Bureau (BR).¹³

5. *EPFD History.* Early proponents of NGSO FSS systems in the 1990s prompted the international community, and the Commission, to develop sharing criteria for NGSO FSS systems to share frequencies with incumbent GSO networks. In 1997, the ITU World Radiocommunication Conference (WRC)¹⁴ first adopted provisional EPFD limits for NGSO FSS systems operating in the Ku-band and Ka-band.¹⁵ These preliminary EPFD limits were studied during the 1998-2000 ITU-R study period.¹⁶ Final EPFD limits were adopted at WRC-2000.¹⁷ The Commission added the ITU's Ku-band EPFD limits to its rules in 2000,¹⁸ incorporated the Ka-band EPFD limits into its rules in 2017,¹⁹ and extended EPFD limits to new NGSO FSS operations in the 17.3-17.8 GHz band in 2024.²⁰

6. In 2019, the international community again considered sharing criteria among GSO and NGSO FSS systems, this time in the higher Q- and V-bands between 37.5 GHz and 51.4 GHz.²¹ WRC-19

¹¹ EPFD downlink limits in these bands apply on both a per-system (single-entry) basis and on an aggregate basis (among all operating NGSO FSS systems). In addition to the downlink (space-to-Earth) limits, EPFD limits in the 10.7-30 GHz range also apply to uplink (Earth-to-space) transmissions and inter-satellite transmissions. International Telecommunication Union (ITU) Radio Regulations Art. 22, Nos. 22.2, 22.5I, Tables 22-1A, 22-1B, 22-1C, 22-1D, 22-2, 22-3, 22-4A, 22-4A1, 22-4B, 22-4C; ITU Resolution 76 (Rev.WRC-15); *see also* 47 CFR § 25.108(c)(3).

¹² 47 CFR §§ 25.114(d)(12), 25.137(c), 25.146(a)(2).

¹³ 47 CFR § 25.146(c); *see also* ITU Resolution 85 (Rev.WRC-23). The BR uses a software tool to determine conformity with EPFD limits. Recommendation ITU-R S.1503 provides a functional description used in the development of this software tool. For the mission of the ITU Radiocommunication Sector (ITU-R), *see* <https://www.itu.int/en/ITU-R/information/Pages/mission-statement.aspx>. Before the availability of the EPFD validation software, the BR had requested commitments from the notifying administrations that they will meet the EPFD limits and, under these commitments, the BR gave "qualified favorable" findings to their systems. *See* Recommendation ITU-R S.1503 (Rev.WRC-23), *considering f*). Some NGSO FSS systems are pending review of the "qualified favorable" findings despite the availability of EPFD validation software. *Id.* at *recognizing*.

¹⁴ WRCs are held every three to four years. A WRC reviews, and, if necessary, revises the ITU Radio Regulations, the international treaty governing the use of the radio-frequency spectrum and the associated geostationary-satellite and non-geostationary-satellite orbits. <https://www.itu.int/en/ITU-R/conferences/wrc/Pages/default.aspx>.

¹⁵ *See Amendment of Parts 2 and 25 of the Commission's Rules to Permit Operation of NGSO FSS Systems Co-Frequency with GSO and Terrestrial Systems in the Ku-Band Frequency Range*, First Report and Order, 16 FCC Rcd 4096, 4106, para. 12 (2000). In this Order and Notice of Proposed Rulemaking, the term "Ku-band" refers to the 10.7-14.5 GHz frequency range, and the term "Ka-band" refers to the 17.3-30 GHz frequency range.

¹⁶ *See id.*, 16 FCC Rcd at 4106-07, paras. 12, 15.

¹⁷ *See id.*, 16 FCC Rcd at 4128, para. 72.

¹⁸ *See id.*, 16 FCC Rcd at 4100, para. 2.

¹⁹ *Update to Parts 2 and 25 Concerning Non-Geostationary, Fixed-Satellite Service Systems and Related Matters*, Report and Order and Further Notice of Proposed Rulemaking, 32 FCC Rcd 7809, 78200-21, 7822-23, paras. 35, 41-42 (2017) (*NGSO FSS Report and Order*); 47 CFR §§ 25.146(a)(2), (c), 25.289.

²⁰ *Amendment of Parts 2 and 25 of the Commission's Rules to Enable NGSO Fixed-Satellite Service (Space-to-Earth) Operations in the 17.3-17.8 GHz Band*, Report and Order, IB Docket No. 22-273, FCC 24-97, para. 28 (2024) (*17 GHz NGSO Order*); 47 CFR § 25.146(a)(2).

²¹ *See* WRC-19 agenda item 1.6.

did not adopt EPFD limits in these bands, however. Instead, it required NGSO FSS systems to protect GSO networks under specific short-term²² and long-term²³ protection criteria that incorporate a degraded throughput methodology.²⁴

7. *Current ITU Work.* WRC-23 considered a proposal from the Inter-American Telecommunication Commission (CITEL)²⁵ co-signed by ten member states, including the United States, to review the EPFD limits under a future agenda item for WRC-27.²⁶ While the proposed agenda item was not adopted, WRC-23 invited ITU-R to conduct technical studies on the EPFD limits and to inform WRC-27 of the results of the studies, without any regulatory consequences.²⁷ These studies are being carried out in ITU-R Working Party (WP) 4A.²⁸

8. *Petition for Rulemaking.* On August 9, 2024, SpaceX filed a petition for rulemaking asking the Commission to revisit the NGSO-GSO sharing regime in downlink frequency bands between 10.7 GHz and 30 GHz that are subject to EPFD limits and to amend sections 25.146 and 25.289 of the Commission's rules.²⁹ SpaceX argues that the EPFD limits referenced in these rule sections are based on "flawed and outdated assumptions"³⁰ that result in inefficient spectrum sharing between GSO and NGSO systems.³¹ SpaceX states that compliance with EPFD limits imposes "wide-ranging constraints and costs on virtually all aspects of next-generation" NGSO FSS systems.³² SpaceX asks that the Commission develop an updated sharing framework, with appropriate short-term and long-term GSO protection criteria and realistic GSO reference links, that NGSO FSS operators could use in the United States as an alternative to compliance with the current EPFD limits in these bands.³³

²² The short-term protection criteria is based on the percentage increase of the time allowance for the carrier-to-noise (C/N) value associated with the shortest percentage of time specified in the short-term performance objective of generic GSO reference links. ITU Radio Regulations, Art. 22, Nos. 22.5L, 22.5M.

²³ The long-term protection criteria is based on the percentage reduction in time-weighted average spectral efficiency calculated on an annual basis for generic GSO reference links using adaptive coding and modulation. ITU Radio Regulations, Art. 22, Nos. 22.5L, 22.5M.

²⁴ ITU Radio Regulations, Art. 22, Nos. 22.5L, 22.5M; ITU Resolution 769 (WRC-19); ITU Resolution 770 (Rev. WRC-23).

²⁵ CITEL is an agency of the Organization of American States (OAS) that works to promote telecommunications and information and communication technologies (ICT) in the Americas. <https://www.oas.org/ext/en/main/oas/our-structure/agencies-and-entities/citel/Home>.

²⁶ CITEL, Proposals for the Work of the Conference (WRC-23), Document WRC23/44/A27/A4 (June 26, 2023) (CITEL WRC-23 Proposal).

²⁷ Minutes of the Eleventh Plenary Meeting of WRC-23, Document 526-E, at 5 (Jan. 15, 2024), stating: "WRC-23 invites ITU-R to conduct technical studies on the epfd limits in Article 22, including the epfd limits referred to in No. 22.5K, in order to ensure the continued protection of GSO FSS and BSS networks, and to inform WRC-27 of the results of the studies, without any regulatory consequences. This work should not be submitted under agenda item 9.1."

²⁸ See WP 4A Chair's Report, Document 4A/343, Annex 20.

²⁹ Petition; 47 CFR §§ 25.146, 25.289.

³⁰ Petition at 10.

³¹ See *id.* at 10, 13-5.

³² Petition at 15.

³³ See *id.* at 20-22.

9. The Commission sought comment on the Petition on October 2, 2024.³⁴ Seven comments, two oppositions, and eight reply comments were filed.³⁵ Most parties support a review of the current EPFD limits.³⁶ Several argue the Commission should initiate a rulemaking now,³⁷ while others suggest the Commission postpone until further ITU developments.³⁸ Viasat, which has previously urged the Commission to rethink the Ka-band EPFD limits,³⁹ opposes the Petition and argues that current EPFD limits must be preserved.⁴⁰ EchoStar cites particular concerns regarding the BSS and terrestrial operations in the 12.2-12.7 GHz band in its opposition to the Petition.⁴¹

III. DISCUSSION

A. SpaceX Petition for Rulemaking

10. Given the Commission's obligation "to make available, so far as possible, to all the people of the United States . . . world-wide wire and radio communication service" and given the slow pace of international progress relative to the rapid sprint at which the American space industry is operating, we recognize that now is the time for us to act to ensure America leads the way to the future of space.⁴² Therefore, after review of the Petition and the comments and oppositions filed, we conclude that the record on the Petition discloses sufficient reasons to justify initiating a rulemaking proceeding seeking further comment on the proposal.⁴³ Indeed, the time is ripe to revisit the NGSO-GSO sharing regime in the 10.7-12.7, 17.3-18.6, and 19.7-20.2 GHz frequency bands developed a quarter century ago.⁴⁴ Modern

³⁴ *Office of the Secretary, Reference Information Center, Petition for Rulemaking Filed*, Public Notice, Report No. 3210 (Oct. 2, 2024).

³⁵ See Appendix B – List of Commenters, *infra*.

³⁶ Kuiper Comments at 3, 6; Telesat Comments at 2; JSX Comments at 1; SpaceX Comments at 1; SES Comments at 3; USAFA Cadets Reply at 1; *see also* Eutelsat Comments at 1-2 (arguing the Commission should "continue working toward international consensus to update the EPFD limits"); DIRECTV Reply at 3 (concurring with Eutelsat); *see also generally* EchoStar Opposition at 2 (arguing "[t]he complex technical and policy issues surrounding EPFD limits demand careful study and international collaboration").

³⁷ See Kuiper Comments at 6; Telesat Comments at 4; JSX Comments at 1; SpaceX Comments at 1.

³⁸ See SES Comments at 11; Eutelsat Comments at 3; DIRECTV Reply at 2-3; USAFA Cadets Reply at 1; *see also* CTIA Comments at 6.

³⁹ See Comments of Viasat, Inc., IB Docket No. 16-408, at 18 (filed Feb. 27, 2017) ("[I]t is imperative that this proceeding evaluate the ITU's EPFD limits anew to determine if they remain appropriate in light of current GSO technology.").

⁴⁰ Viasat Opposition at 2.

⁴¹ See EchoStar Opposition at 13-16, 17-19.

⁴² 47 U.S.C. § 151; *see also* 47 U.S.C. § 157(a) ("It shall be the policy of the United States to encourage the provision of new technologies and services to the public.").

⁴³ See 47 CFR § 1.407; *see also* CITEL WRC-23 Proposal at 2 ("Given the importance of both GSO and non-GSO for FSS implementations to the delivery of numerous types of services and applications, including the delivery of high-speed broadband services to rural and remote areas, it is imperative to ascertain whether existing RR Article 22 epfd limits are suitable to allow for the efficient use of the orbit and spectrum resources[.]"). Thus, we grant SpaceX's request to initiate a rulemaking to "permit more efficient sharing between GSO and NGSO systems in downlink bands between 10.7 GHz and 30 GHz that are subject to EPFD limits." SpaceX Petition at 11. As noted in para. 4 *supra*, in the United States these are the 10.7-12.7, 17.3-18.6, and 19.7-20.2 GHz bands.

⁴⁴ See Working Document Towards A Preliminary Draft New Report [Article 22 Epfd Limit Studies], United States Contribution to Working Party 4A, Document 4A/84, at 1-2 (Apr. 19, 2024) (U.S. WP 4A Contribution) ("Given the advancements in satellite technologies and spectrum management techniques over the last 25 years, the RR Article 22 epfd limits are in need of a fresh technical review to ensure they continue to protect GSO FSS and BSS networks

(continued....)

NGSO satellite constellations delivering high-speed broadband connectivity bear little resemblance to the theoretical NGSO systems considered in the 1990s.⁴⁵ Satellite technologies and spectrum management techniques have evolved.⁴⁶ NGSO modeling has improved.⁴⁷ GSO networks have continued to innovate.⁴⁸

11. Despite the ongoing revolution in the FSS marketplace, compliance with EPFD limits based on 90s-era system designs significantly limit the services offered by NGSO broadband satellite constellations today.⁴⁹ To comply with these limits, NGSO system designers state that they adjust four key operational parameters of their systems, each of which can affect the quality of service experienced by the end user.⁵⁰ First, they limit radiated power levels (effective isotropic radiated power (EIRP)), reducing signal quality on the ground and making it harder to ensure a consistently high level of service.⁵¹ Second, they implement wide “avoidance angles” of the geostationary arc, which can reduce the coverage of the entire system as other satellites are diverted to fill in the gap created by satellites that may not transmit because they fall within the “avoidance angle.”⁵² Third, they restrict the number of satellites simultaneously serving a particular location on the ground (Nco), constraining the number of co-frequency beams that an NGSO system may place in one area and thereby limiting its ability to increase capacity resulting in less efficient spectrum sharing.⁵³ And fourth, they employ higher earth station antenna minimum elevation angles, which, in combination with GSO avoidance angles, reduce the area of the sky above these earth stations that NGSO operators have available, affecting service as well as coordination options with other NGSO operators.⁵⁴ Since EPFD limits must be met at every location on Earth, even locations in which a GSO network has no earth stations present, a percentage of NGSO satellites and their capacity cannot be utilized to serve consumers even where there is no risk of harmful

from unacceptable interference without unnecessarily over constraining non-GSO operations in a way that results in inefficient spectrum sharing between non-GSO FSS and GSO systems to the detriment of non-GSO FSS deployment.”); Petition at 1; Kuiper Comments at 3, 6; Telesat Comments at 2; JSX Comments at 1.

⁴⁵ See Petition at 4, 9; Kuiper Comments at 7-8. While the software tool used by the BR to determine conformity with EPFD limits has been updated through changes to Recommendation ITU-R S.1503, this has not changed the limits themselves. *But see* SES Comments at 6.

⁴⁶ See Petition at 4-5; U.S. WP 4A Submission at 2; *see also, e.g.*, Eutelsat Comments at 2 (noting “changes in GSO and NGSO systems since [EPFD] limits were adopted, reflecting the tremendous growth of the satellite industry”).

⁴⁷ See generally Petition at 9.

⁴⁸ See, e.g., 2024 *Marketplace Report*, FCC 24-136, para. 425; *NGSO FSS Report and Order*, 32 FCC Rcd at 7820, para. 35 (recognizing that Ka-band EPFD limits “were not developed with the most advanced modern GSO networks in mind”).

⁴⁹ See U.S. Contribution to WP 4A at 4 (“[T]he ability to operate and provide services from today’s operational and planned non-GSO systems are significantly penalized when subjected to the limits based on the designs from the time period from before 2000.”).

⁵⁰ See Petition at 15-19; Kuiper Comments at 4-6; *see also* U.S. WP 4A Contribution at 5.

⁵¹ See Petition at 16; Kuiper Comments at 5 *see also* U.S. WP 4A Contribution at 5.

⁵² See Petition at 16, 18; Kuiper Comments at 5 *see also* U.S. WP 4A Contribution at 5-9. For example, an initial study has indicated that reducing the geostationary arc avoidance angle of a hypothetical NGSO system of 425 satellites operating at 700 km from 16 degrees to 6 degrees would cut the loss of global coverage from 30% to 10%. *See* U.S. WP 4A Contribution at 8.

⁵³ See Petition at 17-18; Kuiper Comments at 5 *see also* U.S. WP 4A Contribution at 5.

⁵⁴ Petition at 18-19 *see also* U.S. WP 4A Contribution at 5.

downlink interference to GSO networks.⁵⁵ In total, compliance with the current EPFD limits constrains NGSO systems and directly degrades the efficiency of spectrum use by these systems.⁵⁶

12. Beyond their significant impact on burgeoning NGSO systems, the particular EPFD limits themselves have raised questions as to whether they constitute the most efficient spectrum sharing regime for GSO and NGSO systems in the 10.7-12.7 GHz, 17.3-18.6 GHz, and 19.7-20.2 GHz bands. As an illustrative example in the Ku-band, one initial study has indicated that if GSO satellite operators were required to meet the current NGSO FSS EPFD limits with respect to other GSO operators, it would require an orbital separation of between 25-30 degrees between GSO satellites, which is four-to-five times more restrictive than the six-degree separation that GSO operators have determined would lead to negligible interference internationally, and 12-to-15 times more restrictive than the longstanding two-degree spacing rules the Commission has determined promotes more efficient GSO FSS operations in the United States.⁵⁷ Similarly, in the Ka-band, if the current NGSO EPFD limits that apply in the 19.7-20.2 GHz band had to be met among GSO FSS operators, it could require GSO orbital separations of 46 degrees—more than five times the GSO separation of eight degrees used internationally, and 23 times the GSO separation used for service in the United States.⁵⁸ Potential internal inconsistencies in the NGSO FSS EPFD limits also stand out. In the Ka-band, the NGSO FSS EPFD limit in the upper portion of the band is 15 dB (31.6 times) lower than the NGSO FSS EPFD limit in the lower portion of the band, despite only a 1.1 gigahertz separation between the two.⁵⁹ And the overall methodology used to derive the current NGSO FSS EPFD limits, including whether it incorporated an appropriate long-term protection criteria,⁶⁰ or used unrealistic GSO reference links,⁶¹ has also been questioned by parties.

13. Initiating a review of the current NGSO-GSO sharing regime in the 10.7-12.7 GHz, 17.3-18.6 GHz, and 19.7-20.2 GHz bands in the United States is consistent with longstanding Commission precedent and priorities. Indeed, the cornerstone of Commission policy on GSO-GSO FSS spectrum sharing, premised on two-degree orbital spacing between GSO satellites, resulted from independent Commission consideration and is a departure from more conservative spacings of six degrees or greater

⁵⁵ See Petition at 18; Kuiper Comments at 6.

⁵⁶ See U.S. WP 4A Contribution at 1.

⁵⁷ See U.S. WP 4A Contribution at 15-17; *see, e.g.*, 47 CFR § 25.103 (defining “two-degree-compliant space station” as a GSO FSS space station operating in the conventional or extended C-bands, the conventional or extended Ku-bands, the 24.75-25.25 GHz band, or the conventional or extended Ka-bands within the limits on downlink equivalent isotropically radiated power (EIRP) density or PFD specified in [§ 25.140\(a\)\(3\)](#) or [\(b\)\(3\)](#) and communicating only with earth stations operating in conformance with routine uplink parameters specified in [§ 25.211\(d\)](#), [§ 25.212\(c\)](#), [\(d\)](#), or [\(f\)](#), or [§ 25.218](#)).

⁵⁸ See U.S. WP 4A Contribution at 15-17; *see also, e.g.*, Telesat Reply at 1-2; *but see* SES Comments at 7; EchoStar Reply at 10.

⁵⁹ See U.S. WP 4A Contribution at 10-12; *see also, e.g.*, Comments of WorldVu Satellites Limited, d/b/a OneWeb, IB Docket No. 16-408, at 23-24 (filed Feb. 27, 2017) (arguing the Commission should consider allowing NGSO FSS operators the choice of meeting the Ka-band EPFD limits that apply in the 17.8-18.6 GHz band or the limits that apply in the 19.7-20.2 GHz band for operations in either band, as “[w]hilst this would be a deviation from ITU Article 22, it may provide some flexibility to NGSO FSS systems operating over the U.S., with no corresponding impact on the GSO FSS networks”).

⁶⁰ See CITEL WRC-23 Proposal at 6 (recognizing “that Article 22 and Resolution 76 (Rev.WRC 15) epfd limits were derived taking into account only a short-term protection criterion”); Petition at 5; Kuiper Comments at 8; Telesat Comments at 2; *but see* SES Comments at 6, 8; EchoStar Reply at 10.

⁶¹ *See, e.g.*, SpaceX Comments at 3, 5-6; *see also* U.S. Contribution to WP 4A at 3 (noting that some “power levels and noise temperatures used in the reference links do not appear to be consistent with typical GSO operations”); *but see* SES Reply at 5-6.

applied internationally.⁶² Our closer-spacing allowances for GSO FSS satellites have enabled more satellites to serve the United States for decades,⁶³ enhancing competition and service offerings for American consumers. More recently, the Commission has adopted clear rules for NGSO-NGSO FSS spectrum sharing while comparable international guidance remains long under development.⁶⁴ Even when adopting the current Ka-band EPFD limits into our rules, the Commission noted that they “were not developed with the most advanced modern GSO networks in mind” but it concluded that, in the absence of any alternative limits proposed on the record, “it would not be advisable to remain without Ka-band EPFD limits in our rules pending such deliberations.”⁶⁵ Similarly, last year when the Commission adopted EPFD limits on new NGSO FSS operation in the 17.3-17.8 GHz band, consistent with decisions taken at WRC-23, the Commission noted international efforts to modernize EPFD limits and stated that it may reevaluate its rules considering international technical developments.⁶⁶ This rulemaking will provide a dedicated forum for the Commission to reassess the complex technical issues associated with NGSO-GSO spectrum sharing in the 10.7-12.7, 17.3-18.6, and 19.7-20.2 GHz bands, including any international technical developments.

14. The status of international work in this area also supports a review by the Commission for services in the United States of the current NGSO-GSO sharing regime reflected in EPFD limits. ITU-R studies are ongoing, as they have been for years. There is no explicit item on the agenda for WRC-27 to review EPFD limits,⁶⁷ and the invitation for ITU-R study of the matter during the current study period includes the phrase “without any regulatory consequences.”⁶⁸ Initiating our review of the NGSO-GSO sharing regime now will enable us to have a full record to make any needed changes to our rules for the benefits of any revised NGSO-GSO sharing regime to American consumers in a timely manner.⁶⁹ While we move forward on behalf of consumers in the United States, we will continue to monitor and participate in international studies and share our experience with other countries. We

⁶² See, e.g., *Licensing Space Stations in the Domestic Fixed-Satellite Service*, Report and Order, 48 F.R. 40233 (Sept. 6, 1983).

⁶³ See, e.g. *Comprehensive Review of Licensing and Operating Rules for Satellite Services*, Second Report and Order, 30 FCC Rcd 14713, 14747, para. 92 (2015) (noting the Commission’s “two-degree spacing policy was adopted to increase, to the maximum feasible extent, the number of orbital locations from which GSO FSS space stations can provide service in the United States”).

⁶⁴ See *Revising Spectrum Sharing Rules for Non-Geostationary Orbit, Fixed-Satellite Service Systems*, Second Report and Order and Order on Reconsideration, FCC 24-117 (2024); *Revising Spectrum Sharing Rules for Non-Geostationary Orbit, Fixed-Satellite Service Systems*, Report and Order and Further Notice of Proposed Rulemaking, 38 FCC Rcd 3699 (2023); *NGSO FSS Report and Order*, 32 FCC Rcd 7809; see also WP 4A Chair’s Report, Document 4A/343, Annex 30.

⁶⁵ *NGSO FSS Report and Order*, 32 FCC Rcd at 7820, para. 35.

⁶⁶ See *17 GHz NGSO Order*, FCC 24-97, para. 23 n.91. There is also no inconsistency with the Commission’s decision not to develop *ad hoc* EPFD limits in the context of a satellite licensing proceeding, when work under an established WRC agenda item indicated that EPFD limits may not be necessary and when alternative NGSO-GSO sharing criteria were adopted by the WRC little over a year later. *But see* Viasat Opposition at 9, citing *Space Exploration Holdings, LLC, Application for Approval for Orbital Deployment and Operating Authority for the SpaceX V-band NGSO Satellite System*, Memorandum Opinion, Order and Authorization, 33 FCC Rcd 11434, 11439, para. 10 (2018).

⁶⁷ ITU Resolution 813 (WRC-23).

⁶⁸ Minutes of the Eleventh Plenary Meeting of WRC-23, Document 526-E, at 5 (Jan. 15, 2024).

⁶⁹ See generally Kuiper Comments at 3 (“[T]he Commission need not wait for the ITU to act because the Radio Regulations specifically allow for countries to adopt different limits within their own territory. There is therefore no need for U.S. businesses and customers to wait to reap the benefits of common-sense rule updates.”); SpaceX Comments at 1; Telesat Comments at 2; JSX Comments at 1; *but see* SES Comments at 8-9; Eutelsat Comments at 3; DIRECTV Reply at 2-3; USAFA Cadets Reply at 1; *but see also* CTIA Comments at 6.

anticipate a Commission rulemaking may spur additional discussion and technical analyses that benefit international deliberations as well.⁷⁰ We are happy to lead in this way and encourage the international community to modernize their domestic as well as international frameworks. As we start this proceeding, however, our primary focus will be on developing a domestic record that will enable the Commission to make an informed decision with the best interests of American consumers in mind.⁷¹ International updates can be done in due course.⁷²

15. While the suitability of existing NGSO FSS EPFD limits to the modern satellite environment is therefore in need of a fresh review, the fundamental importance of ensuring compatibility between NGSO FSS systems and GSO FSS and BSS networks remains unchanged. Modern satellites in GSO and NGSO orbits are helping bridge the digital divide across the United States, particularly in the most rural areas of the country where it may be uneconomical to build terrestrial networks.⁷³ GSO and NGSO FSS satellites currently under construction are expected to offer broadband connectivity at fiber-like speeds.⁷⁴ At the same time, GSO BSS satellites continue to serve millions of satellite television subscribers.⁷⁵

16. GSO and NGSO satellites also coexist in shared and adjacent spectrum environments with other services, including terrestrial and radio astronomy services.⁷⁶ This proceeding will not change the existing PFD limits that apply to GSO and NGSO systems to protect terrestrial services.⁷⁷ However, we will consider how any modifications to the NGSO-GSO sharing regime might affect terrestrial and radio astronomy stations.

17. While NGSO-GSO spectrum sharing conditions vary in different portions of the 10.7-12.7 GHz, 17.3-18.6 GHz, and 19.7-20.2 GHz bands, this proceeding will provide a single U.S. forum for the complex technical discussions that underlie NGSO-GSO sharing in all of the bands, as well as flexibility to consider operations in different frequency segments individually. We believe that initiating multiple NGSO-GSO sharing proceedings based on different frequency bands would risk multiplication of the substantial technical work and discussions necessary to evaluate this difficult issue.⁷⁸ Particular concerns associated with any potential alternative frameworks will be considered based on this record.

⁷⁰ See, e.g., Telesat Comments at 3 (noting that while the “lengthy international effort” is ongoing, “the FCC has an opportunity to provide strong leadership to drive this complicated issue forward by providing a forum to focus the discussion”).

⁷¹ See, e.g., SpaceX Reply at 5 (“By tailoring GSO-NGSO sharing rules based solely on reference GSO links operating over the United States, the Commission can achieve an even more efficient framework than a potential new ITU framework.”). For the reasons stated in this paragraph, we disagree with commenters who suggest we delay initiating a rulemaking proceeding pending further ITU developments. See SES Comments at 8-9; Eutelsat Comments at 3; DIRECTV Reply at 2-3; USAFA Cadets Reply at 1; see also CTIA Comments at 6.

⁷² See, e.g., Telesat Comments at 4 (“There is no point in clinging to an outdated regime when a better approach presents itself; [international] harmonization can always occur after the FCC updates its rules.”); but see, e.g., Eutelsat Comments at 6, 7 and DIRECTV Reply at 3-4 (generally arguing the Commission should defer rulemaking for the sake of international harmonization).

⁷³ 2024 Marketplace Report, FCC 24-136, para. 185.

⁷⁴ *Id.*, para. 194.

⁷⁵ See *id.*, para. 204.

⁷⁶ See 47 CFR §§ 1.924, 2.106(a), (c)(131).

⁷⁷ 47 CFR §§ 25.146(a)(1), 25.208.

⁷⁸ For these reasons we disagree with commenters who urge us to proceed only on a band-by-band or service-by-service basis. See CTIA Comments at 8; DIRECTV Reply at 4-5.

18. In light of the above, the oppositions to the Petition do not justify its dismissal or denial. Contrary to Viasat's claims, initiating a domestic review of the NGSO-GSO sharing regime is fully consistent with Commission precedent promoting efficient spectrum sharing in the United States.⁷⁹ EchoStar's objections do not counsel against initiating a rulemaking given that it would have an opportunity to submit comments and contribute to developing the technical record in this rulemaking. Domestic rulemaking will enhance, not "jeopardize," ongoing international discussions.⁸⁰ Developing a technical record in this proceeding, without any proposed rules, also does not constitute "rash action."⁸¹ There is no inconsistency between the Commission's prior decision not to have its staff duplicate the time-consuming technical review of EPFD showings that is already performed by the staff of the ITU BR, using ITU-approved software, and our decision to revisit the sharing regime itself, which could substantially improve efficiency and competition.⁸² Nor will including the 12.2-12.7 GHz band in this rulemaking prejudge the outcome of the separate 12 GHz band rulemaking; on the contrary, excluding the band here would forgo the important technical record to be developed in this proceeding.⁸³ Finally, EchoStar's assertion that derogation from international EPFD limits "prior to an ITU consensus would also result in harmful interference and disruption of service" cannot simply be accepted. Such a claim must specify in detail what the change would be, and the Commission must analyze these claims. However, in any event, the concerns EchoStar raises can be fully evaluated in the context of this rulemaking prior to any decision on a potential alternative NGSO-GSO sharing framework.⁸⁴ Accordingly, we deny the oppositions to initiating a rulemaking on these issues.

B. Notice of Proposed Rulemaking

19. *Overview.* This Notice of Proposed Rulemaking seeks to develop a substantial technical record concerning modern and efficient spectrum sharing among NGSO FSS systems and GSO FSS and BSS networks in the 10.7-12.7, 17.3-18.6, and 19.7-20.2 GHz bands, while ensuring that any rule changes do not affect the continued protection of co-frequency terrestrial services. We seek comment on how the satellite industry has changed since ITU EPFD limits were adopted in 2000; how the current limits protect GSO networks and impact NGSO systems; what a modern, realistic set of GSO reference links would be; what short-term and long-term GSO protection criteria would be appropriate; methodological considerations; alternative NGSO-GSO sharing frameworks; information sharing; aggregate interference;

⁷⁹ See Viasat Opposition at 3-5, 9. Though Viasat also argues there is no "pressing need" to initiate a rulemaking given that NGSO FSS systems are able to provide "high-quality satellite broadband services" while operating within EPFD limits, the potential for faster satellite broadband, better coverage, and more efficient competition clearly warrant our review in this case. *Id.* at 9. In addition, noting that ITU-R is currently performing technical studies on the EPFD limits per an invitation from WRC-23, our domestic consideration of these limits does not "contradict" decisions at WRC-23. *Id.* at 2.

⁸⁰ EchoStar Opposition at 4.

⁸¹ See *id.* at 5-7.

⁸² See *id.* at 7-9. Additionally, while the question of any ultimate revision to the NGSO-GSO sharing regime in the 10.7-12.7 GHz, 17.3-18.6 GHz, and 19.7-20.2 GHz bands in the United States will be made later on the basis of the record in this proceeding, it is beyond question that the United States has the sovereign right to regulate telecommunications within its territory, including by permitting EPFD levels in excess of international limits. See, e.g., Constitution of the International Telecommunication Union, Preamble ("fully recognizing the sovereign right of each State to regulate its telecommunication"); ITU Radio Regulations, No. 22.5CA (noting EPFD limits "may be exceeded on the territory of any country whose administration has so agreed"); but see EchoStar Opposition at 9 (alleging that Commission action to alter EPFD limits in the United States would "turn the Commission into an insurrectionist iconoclast operating outside the bounds of international law").

⁸³ See *Expanding Flexible Use of the 12.2-12.7 GHz Band*, WT Docket No. 20-443, Report and Order and Further Notice of Proposed Rulemaking, 38 FCC Rcd 5283, 5284, para. 2 (2023) (further investigat[ing] the potential to expand terrestrial fixed use or to permit unlicensed use . . .).

⁸⁴ EchoStar Opposition at 13; see *id.* at 12-19; EchoStar Reply, Exh. 1.

protection of terrestrial stations and radio astronomy; cross-border considerations; compliance; costs and benefits; and transitional measures.⁸⁵ A robust record developed from these inquiries will assist the Commission in promoting a more competitive and efficient marketplace for satellite broadband and other services in these bands.

20. *Changes since 2000.* As an initial matter, we invite comment on what relevant changes have taken place in the satellite industry since the adoption of EPFD limits by the ITU in 2000. How have the assumptions underlying this NGSO-GSO spectrum sharing regime changed? How have the capabilities and techniques of current and future GSO networks changed regarding their ability to share spectrum? How has NGSO system design changed? Are the modeling assumptions used in the verification of compliance with EPFD limits, such as continuous transmission at maximum power or the use of a worst-case geometry algorithm, realistic with respect to current NGSO operations? Are the protection criteria (e.g., I/N ratios) assumptions still valid or do current satellite technology and design warrant different criteria?

21. *Current EPFD Protection of GSOs.* Next, we invite comment on whether the current NGSO EPFD limits may overprotect modern GSO networks. What levels of protection (short-term and long-term) do GSO networks reasonably require? Are the current EPFD limits appropriate for those requirements? What levels of protection do the current EPFD limits offer to GSO networks? If a modern NGSO system is compliant with the EPFD limits, what is the expected loss in throughput or increase in unavailability for a modern GSO satellite link that relies on Adaptive Coding and Modulation (ACM)?⁸⁶ Does this expected level of degradation vary for gateways and user terminal earth stations?⁸⁷ What is the level of short-term protection provided by current NGSO EPFD limits to GSO satellite links that do not rely on ACM? How do these levels of protection compare to other sources of interference, such as attenuation due to atmospheric conditions such as rain or interference from other GSO networks? How do these levels of protection compare to any triggers for coordination? Are the protection criteria on which the EPFD limits are based (e.g., I/N ratios) still valid?

22. *Current Impact on NGSOs.* At the same time, do the current EPFD limits overly restrict modern NGSO operations? What are the impacts of the operational changes that NGSO FSS systems make to comply with the current EPFD limits on the quality or availability of service within the United States? To what extent would NGSO systems be able to offer higher capacity or better service in certain areas with less restrictive limits? We encourage commenters to submit studies showing the impact of less restrictive limits on capacity and other measures of service quality.

23. What GSO arc avoidance angles, or restrictions on where NGSO FSS systems can steer their antenna beams to prevent in-line interference events with GSO satellite networks, are NGSO operators adopting to comply with current EPFD limits, and how do those differ from the angles they could operate with less restrictive limits? How do they differ from the equivalent separation angles used by GSOs for inter-arc sharing? How much must NGSO operators restrict their power levels to ensure

⁸⁵ See generally Kuiper January 31, 2025 *ex parte* at 2-6.

⁸⁶ ACM or Link adaptation is a term used in wireless communications to denote the match of the modulation, coding, and other signal and protocol parameters to the conditions on the radio link (e.g., the pathloss, interference due to signals coming from other transmitters, the sensitivity of the receiver, the available transmitter power margin, etc.). For example, [WiMAX](#) uses a rate adaptation algorithm that adapts the modulation and coding scheme (MCS) according to the quality of the radio channel, and thus the bit rate and robustness of data transmission. Shami, Abdallah; Maier, Martin; Assi, Chadi (2010-01-23). [Broadband Access Networks: Technologies and Deployments](#). Springer Science & Business Media, at 100.

⁸⁷ A gateway earth station is an earth station, or complex of multiple earth station antennas, that supports the routing and switching functions of a satellite system and that does not originate or terminate communication traffic. See 47 CFR § 25.103 (defining NGSO FSS gateway earth station). User terminals typically employ smaller antennas than gateways.

compliance with EPFD limits and how do they implement such power level changes? Do they need to restrict their power levels differently in different bands given the EPFD limits? What Nco, representing the number of co-frequency, co-polarization satellite beams transmitting to a given point on the Earth's surface simultaneously, must NGSO operators adopt to comply with the EPFD limits? What Nco could NGSO operators adopt if these limits were not in place, and how would that change their available capacity? Given how the EPFD limits were designed, how well do modern NGSO operations map onto the EPFD limits? Is it plausible for a modern NGSO operator to take full advantage of the required masks, or are the masks more restrictive than they appear? How do these levels of restriction differ in the Ku-band and the Ka-band?

24. *GSO Reference Links.* We also seek input on developing a set of modern GSO reference links to be used for analysis. What are the most representative GSO links used in the United States today? What is an appropriate set of GSO FSS reference links in these bands? What is an appropriate set of GSO BSS reference links in these bands? Which of these links use ACM, and which do not? What is the actual antenna pattern performance for these GSO earth stations? How many locations should be used in analyzing GSO reference links, and how can the Commission ensure that those analyzed appropriately reflect the impact of differing geographies on performance? What would be appropriate rain and interference margin for proposed links, and what would be their C/N below which the GSO link is lost? What is the receive system noise temperature of these reference links? What criteria should be used for evaluating the potential for harmful interference? What, if anything, should be apportioned for margin above that criteria? Are there any additional costs associated with developing the GSO reference links? If so, we seek estimates of these costs.

25. *Protection of GSOs that Use ACM.* What is the appropriate methodology for protecting GSO operations that use ACM? In the NGSO-NGSO sharing context, the Commission has made use of a degraded throughput methodology, with a long-term protection criterion representing a percentage of time-weighted average throughput degradation and a short-term protection criterion representing a percentage of absolute increase in link unavailability.⁸⁸ A degraded throughput methodology could be implemented by requiring NGSO FSS operators to demonstrate that they will comply with short-term and long-term protection criteria for GSO FSS and GSO BSS networks.⁸⁹ Is degraded throughput the most appropriate methodology to use in the context of NGSO-GSO sharing when the GSO satellite uses ACM? If so, what are the appropriate protection criteria? Is there any reason to think those criteria should be higher or lower, or should differ between gateways and user terminals or based on whether the earth station is using the Ku-band, the lower Ka-band, or the upper Ka-band? Is there a better methodology available to establish the actual impact an NGSO operator is likely to have on a GSO link that uses ACM? In establishing a methodology and thresholds, how should the Commission balance the need for protection of GSO incumbents with maximizing the benefits of NGSO systems? Should the Commission provide NGSO operators with the same flexibility it did in the NGSO-NGSO sharing context to adjust their operations to meet the new methodology and thresholds?

⁸⁸ See *Revising Spectrum Sharing Rules for Non-Geostationary Orbit, Fixed-Satellite Service Systems*, Second Report and Order and Order on Reconsideration, IB Docket No. 21-456, FCC 24-117, para. 9 (2024). The long-term protection criterion compares throughput under time-varying propagation conditions to throughput under time-varying propagation conditions with interference. The calculation of spectral efficiency is focused on satellite systems utilizing ACM by calculating the throughput degradation as a function of C/N, which varies depending on the propagation and interference impacts on the satellite link over the long term. The short-term protection criterion compares link availability under time-varying propagation conditions to link availability under time-varying propagation conditions with interference. *Id.*, n.39.

⁸⁹ See *generally id.* at paras. 9-108 (adopting an NGSO-NGSO FSS sharing framework incorporating a degraded throughput methodology and addressing various issues related to protection levels and demonstrations of compliance); see also *generally* ITU Radio Regulations, Art. 22, No. 22.5L.

26. *Protection of GSOs that Do not Use ACM.* What is the appropriate methodology for protecting GSO operations that do not use ACM? Is a degraded throughput methodology appropriate for such links, and if so, what changes would be needed to the methodology to account for the lack of ACM? Should the only criterion be a short-term protection criterion? If so, what would an appropriate short-term protection criterion be? Alternatively, would an interference-to-noise (I/N) threshold be a more appropriate long-term protection criterion for GSO links that do not use ACM? If the Commission were to adopt an I/N threshold, what should it be and how can the Commission ensure that such a threshold reflects the necessary protection requirements of GSO operations? If the Commission were to adopt an I/N threshold, what should be the associated percentage of time? If the Commission were to adopt such a long-term threshold, what should the short-term protection criterion be for such links, if any? Finally, what is the approximate number and percentage of FSS, BSS GSO satellites that currently do not use ACM?

27. *Methodology.* What modeling assumptions should underlie any methodology? For example, should the Commission adopt a minimum link availability threshold to guard against risk of low-performing links? Should the Commission adopt standardized reference C/N threshold values to account for the performance of efficient, modern modems and receivers? Should the Commission adopt a minimum receiver performance standard, such as a limit on the receiver noise temperature? Should the Commission incorporate interference from other GSO operations into the baseline calculations for any methodology? What is the level of static interference that a GSO network should accept from its neighboring GSO satellites? Should the Commission require the use of particular rain-fade characteristics, such as those found in Recommendations ITU-R P.618 or P.676?

28. *Alternative NGSO-GSO Sharing Frameworks.* More generally, we seek comment on what NGSO-GSO spectrum sharing framework should apply in the 10.7-12.7 GHz, 17.3-18.6 GHz, and 19.7-20.2 GHz bands. Should the Commission revise the current spectrum sharing methodology, which relies on EPFD limits, or develop an alternative methodology that is based on other protection criteria? Should the Commission require GSO and NGSO operators to coordinate in good faith in these bands? Should the Commission continue to require NGSO operators not to cause unacceptable interference into GSO networks, absent coordination? Should the Commission instead require GSO operators to not cause unacceptable interference into NGSO systems in these bands, absent coordination, as the Commission requires in the 18.8-19.3 GHz downlink band,⁹⁰ and what would be the benefits and costs of this alternative to, both, GSO and NGSO operations? Should any protections offered to GSO operations be uniform in scope, or should they be tailored to the actual usage in particular orbital slots? Should any protections offered to GSO operations be the same for earth stations in motion as those offered to fixed earth stations?

29. *Information Sharing.* What information, if any, should GSO operators be required to share to facilitate efficient spectrum use? Should GSO operators be required to share the actual antenna patterns of their deployed earth stations to the extent they differ from those in public records? Should GSO operators be required to share their actual power levels or other system characteristics that could impact their baseline operations? Should GSO operators be required to share the actual spectrum used or spectrum usage patterns of particular satellites to maximize spectral efficiency? Should GSO operators be required to identify links that require particular interference protection as well as the basis for any such heightened need? Should GSO operators be required to share whether their networks rely on ACM and, if not, the link budgets they use? To the extent a GSO network has blanket-authorized earth stations (such as user terminals), should a GSO operator be required to share whether any such stations are in a given area (so as to avoid protecting stations that do not exist)? What confidentiality measures may be needed to facilitate information sharing when viewing potentially commercially sensitive operational data? What are the estimated costs associated with information sharing?

⁹⁰ 47 CFR § 2.106(a), (d)(165).

30. *Aggregate Interference.* How should we address the potential for aggregate interference from multiple NGSO systems into a GSO network? Should we continue to rely on an approach similar to ITU-R Resolution 76 or ITU-R Resolution 769? How should we address any implementation issues regarding aggregate limits? Is an aggregate limit on NGSO-GSO interference in fact necessary to prevent harmful interference, or might the cost of such regulation outweigh its benefits and create an unjustified obstacle to NGSO deployment? In the context of NGSO-NGSO spectrum sharing, the Commission declined to create an aggregate interference threshold.⁹¹ Should we follow a similar approach here? Should we rely on coordination among NGSO and GSO operators to resolve potential issues of aggregate interference?

31. *Protection of Terrestrial Operations.* How would any changes to the NGSO-GSO sharing regime affect co-frequency terrestrial operations? If current EPFD limits are not met, would any additional constraints on NGSO FSS operation be necessary to protect terrestrial wireless operations, beyond the existing PFD limits? How should the Commission ensure that terrestrial services in different portions of the 10.7-12.7, 17.3-18.6, and 19.7-20.2 GHz bands are not adversely impacted?

32. *Radio Astronomy.* We note the extensive recent record of successful coordination and cooperation between radio astronomy systems and satellite systems, facilitated in part by previous FCC rulemakings. Would any additional considerations with respect to NGSO FSS operation be necessary to protect radio astronomy operations, beyond the existing PFD limits? How would any changes to the NGSO-GSO sharing regime affect co-frequency or adjacent-band radio astronomy operations, and how should the Commission ensure that successful coordination continue?

33. *Cross-Border Considerations.* We invite comments on cross border considerations and any safeguards that may be needed to protect GSO operations in neighboring countries. We seek comment on any needed bilateral engagements to alleviate cross-border interference concerns in our region and share our experience, the record of this proceeding, and our findings. How should the Commission coordinate operations near the borders of Canada and Mexico, and how close to the border would NGSO operations need to be to qualify for such coordination? Are there any other jurisdictions that the Commission would need to coordinate with? We will continue to work with our regional and international partners and participate in international studies in various venues. We believe that this rulemaking may spur additional discussion and technical analyses that benefit the international deliberations. With respect to cross border considerations, commenters should identify the concerns and provide qualitative and quantitative analysis of their proposals.

34. *Compliance.* How should the Commission assess compliance with any new NGSO-GSO sharing framework? Would such a framework require additional monitoring and enforcement on part of the Commission? What additional compliance costs would be associated with such monitoring and enforcement? Should NGSO operators be required to demonstrate compliance in their initial application or should authorization be conditional on later showings of compliance? Should NGSO operators be allowed to use information gathered through coordination discussions with other operators, or required to use public information? Are there any incremental costs that NGSO and GSO operators may incur with a new NGSO-GSO sharing framework that moves away from requiring NGSO operators receiving a “favorable” or “qualified favorable” finding by the ITU BR prior to initiating service? Are confidentiality protections needed for such filings, for example if they rely on proprietary or competitively sensitive information?

35. *Costs and Benefits.* We seek specific information on the benefits and costs associated with any proposals for alternative NGSO-GSO sharing frameworks. What are the economic benefits of less restrictive limits on NGSO operations for U.S. consumers and the aggregate economy of the United States? What costs could less restrictive limits potentially impose on GSO services provided to U.S. consumers or for terrestrial services in the United States? Would less restrictive limits lead to increased

⁹¹ *Id.*, FCC 24-117, para. 61.

compliance and monitoring costs near the borders of Canada and Mexico? Would an alternative NGSO-GSO sharing framework incur additional compliance and monitoring costs? We also seek comment on any additional costs or benefits not outlined above.

36. *Transition to New Rules.* Finally, how should we transition to any new NGSO-GSO sharing framework in the United States, if adopted? What time frame(s) would be reasonable? Under what circumstances would an immediate transition be appropriate? Should the Commission sunset any protections for inefficient or outdated technologies or services? Are there any particular technologies or services that would require a longer period of protection under the current EPFD limits? How should we address existing license conditions that may conflict with any new rules we adopt? Should we offer current licensees and market access grantees a simple process to request modification of their authorizations to conform with any new rules, for example via letter request?

IV. PROCEDURAL MATTERS

37. *Ex Parte Rules.* The proceeding this Notice initiates shall be treated as a “permit-but-disclose” proceeding in accordance with the Commission’s *ex parte* rules.⁹² Persons making *ex parte* presentations must file a copy of any written presentation or a memorandum summarizing any oral presentation within two business days after the presentation (unless a different deadline applicable to the Sunshine period applies). Persons making oral *ex parte* presentations are reminded that memoranda summarizing the presentation must (1) list all persons attending or otherwise participating in the meeting at which the *ex parte* presentation was made, and (2) summarize all data presented and arguments made during the presentation. If the presentation consisted in whole or in part of the presentation of data or arguments already reflected in the presenter’s written comments, memoranda or other filings in the proceeding, the presenter may provide citations to such data or arguments in his or her prior comments, memoranda, or other filings (specifying the relevant page and/or paragraph numbers where such data or arguments can be found) in lieu of summarizing them in the memorandum. Documents shown or given to Commission staff during *ex parte* meetings are deemed to be written *ex parte* presentations and must be filed consistent with rule 1.1206(b). In proceedings governed by rule 1.49(f) or for which the Commission has made available a method of electronic filing, written *ex parte* presentations and memoranda summarizing oral *ex parte* presentations, and all attachments thereto, must be filed through the electronic comment filing system available for that proceeding, and must be filed in their native format (e.g., .doc, .xml, .ppt, searchable .pdf). Participants in this proceeding should familiarize themselves with the Commission’s *ex parte* rules.

38. *Filing Requirements.* Pursuant to sections 1.415 and 1.419 of the Commission’s rules, 47 CFR §§ 1.415, 1.419, interested parties may file comments and reply comments on or before the dates indicated on the first page of this document. Comments may be filed using the Commission’s Electronic Comment Filing System (ECFS). See *Electronic Filing of Documents in Rulemaking Proceedings*, 63 FR 24121 (1998).

- *Electronic Filers:* Comments may be filed electronically using the Internet by accessing the ECFS: <http://www.fcc.gov/ecfs>.
- *Paper Filers:* Parties who choose to file by paper must file an original and one copy of each filing.
 - Filings can be sent by commercial overnight courier, or by first-class or overnight U.S. Postal Service mail. All filings must be addressed to the Commission’s Secretary, Office of the Secretary, Federal Communications Commission.
 - Hand-delivered or messenger-delivered paper filings for the Commission’s Secretary are accepted between 8:00 a.m. and 4:00 p.m. by the FCC’s mailing contractor at 9050

⁹² 47 CFR §§ 1.1200 *et seq.*

Junction Drive, Annapolis Junction, MD 20701. All hand deliveries must be held together with rubber bands or fasteners. Any envelopes and boxes must be disposed of before entering the building.

- Commercial courier deliveries (any deliveries not by the U.S. Postal Service) must be sent to 9050 Junction Drive, Annapolis Junction, MD 20701.
- Filings sent by U.S. Postal Service First-Class Mail, Priority Mail, and Priority Mail Express, must be sent to 45 L Street NE Washington, DC 20554.

39. *People with Disabilities:* To request materials in accessible formats for people with disabilities (braille, large print, electronic files, audio format), send an e-mail to fcc504@fcc.gov or call the Consumer & Governmental Affairs Bureau at 202-418-0530 (voice), 202-418-0432 (TTY).

40. *Regulatory Flexibility Act.* The Regulatory Flexibility Act of 1980, as amended (RFA),⁹³ requires that an agency prepare a regulatory flexibility analysis for notice-and-comment rulemaking proceedings, unless the agency certifies that “the rule will not, if promulgated, have a significant economic impact on a substantial number of small entities.”⁹⁴ Accordingly, the Commission has prepared an Initial Regulatory Flexibility Analysis (IRFA) concerning the possible impact of potential rule and policy changes contained in this Notice of Proposed Rulemaking on small entities. The IRFA is set forth in Appendix A. The Commission invites the general public, in particular small businesses, to comment on the IRFA. Comments must be filed by the deadlines for comments on indicated on the first page of this document and must have a separate and distinct heading designating them as responses to the IRFA.

41. *Initial Paperwork Reduction Act Analysis.* This document does not contain proposed information collections subject to the Paperwork Reduction Act of 1995 (PRA), Public Law 104-13. In addition, therefore, it does not contain any new or modified information collection burden for small business concerns with fewer than 25 employees, pursuant to the Small Business Paperwork Relief Act of 2002, Public Law 107-198, see 44 U.S.C. 3506(c)(4).

42. *Providing Accountability Through Transparency Act.* Consistent with the Providing Accountability Through Transparency Act, Public Law 118-9, a summary of this document will be available on <https://www.fcc.gov/proposed-rulemakings>.

V. ORDERING CLAUSES

43. Accordingly, IT IS ORDERED, pursuant to section 1.407 of the Commission’s rules, 47 CFR § 1.407, that the petition for rulemaking filed by Space Exploration Holdings, LLC, Revision of the Commission’s Rules to Establish More Efficient Spectrum Sharing between NGSO and GSO Satellite Systems, RM-11990, IS GRANTED, and the oppositions filed by Viasat, Inc., and EchoStar Corporation ARE DENIED.

44. IT IS FURTHER ORDERED, pursuant to sections 4(i), 7(a), 303, 308(b), and 316 of the Communications Act of 1934, as amended, 47 U.S.C. §§ 154(i), 157(a), 303, 308(b), 316, that this Notice of Proposed Rulemaking IS ADOPTED.

45. IT IS FURTHER ORDERED that the Commission’s Office of the Secretary, SHALL SEND a copy of this Notice of Proposed Rulemaking, including the Initial Regulatory Flexibility Analysis, to the Chief Counsel for Advocacy of the Small Business Administration.

FEDERAL COMMUNICATIONS COMMISSION

⁹³ 5 U.S.C. §§ 601 *et seq.*, as amended by the Small Business Regulatory Enforcement and Fairness Act (SBREFA), Pub. L. No. 104-121, 110 Stat. 847 (1996).

⁹⁴ *Id.* § 605(b).

Marlene H. Dortch
Secretary

APPENDIX A

Initial Regulatory Flexibility Analysis

1. As required by the Regulatory Flexibility Act (RFA),¹ the Federal Communications Commission has prepared this Initial Regulatory Flexibility Analysis (IRFA) of the possible significant economic impact on a substantial number of small entities by the policies and rules proposed in the *Notice of Proposed Rulemaking (NPRM)*. The Commission requests written public comments on this IRFA. Comments must be identified as responses to the IRFA and must be filed by the deadlines provided on the first page of the *NPRM*. The Commission will send a copy of the *NPRM*, including this IRFA, to the Chief Counsel for Advocacy of the Small Business Administration (SBA).² In addition, the *NPRM* and IRFA (or summaries thereof) will be published in the Federal Register.³

A. Need for, and Objectives of, the Proposed Rules

2. The *NPRM* launches a much needed review of the long-standing spectrum sharing regime between geostationary (GSO) and non-geostationary (NGSO) satellite systems operating in the 10.7-12.7, 17.3-18.6, and 19.7-20.2 GHz bands. The decades-old spectrum sharing regime constitutes the primary restrictive regulatory requirement on NGSO satellite systems currently deploying at breakneck speed. Innovation in the satellite industry has witnessed new NGSO satellite operators launching thousands of satellites in the short span of a few years,⁴ and these operators are beginning to offer high-speed, low-latency broadband services.⁵ The industry has also seen existing GSO satellite operators are continuing to deploy powerful, new satellites with enhanced capabilities.⁶ As innovation continues, the Commission's rules must be at the forefront of the quickly evolving space industry, encourage investment, and further the objectives of space leadership by the United States. By taking a fresh look at today's satellite technology and operations, the *NPRM* will support a more efficient and competitive market for satellite broadband and other in-demand services while uncapping the potential of satellite constellations that were unthinkable when the current regime was developed, to the ultimate benefit of American consumers.

3. The *NPRM* seeks comment to develop a substantial technical record concerning modern and efficient spectrum sharing among NGSO FSS (Fixed Satellite Service) systems, GSO FSS, and BSS (Broadcast Satellite Service) networks in the 10.7-12.7, 17.3-18.6, and 19.7-20.2 GHz bands, while ensuring that any rule changes continue to safeguard and maintain the protection of co-frequency terrestrial services. More specifically, the *NPRM* seeks comment on how the satellite industry has changed since International Telecommunication Union (ITU) equivalent power-flux density (EPFD) limits were adopted in 2000, how the current limits protect GSO networks and impact NGSO systems, what a modern, realistic set of GSO reference links would be, what short-term and long-term GSO protection criteria would be appropriate, methodological considerations, alternative NGSO-GSO sharing frameworks, information sharing, aggregate interference, compliance, protection of terrestrial stations, cross-border considerations, costs and benefits, and transitional measures. A robust record developed from these inquiries will assist the Commission in promoting a more competitive and efficient satellite marketplace that can deliver more high-speed broadband services and choices to the American public.

¹ 5 U.S.C. §§ 601 *et seq.*, as amended by the Small Business Regulatory Enforcement Fairness Act (SBREFA), Pub. L. No. 104-121, Title II, 110 Stat. 847 (1996).

² 5 U.S.C. § 603(a).

³ *Id.*

⁴ See *Communications Marketplace Report*, FCC 24-136, paras. 184, 194, 425 (2024) (*2024 Marketplace Report*).

⁵ See *id.*, paras. 179, 184.

⁶ See *id.*, paras. 184, 194, 425.

B. Legal Basis

4. The proposed action is authorized pursuant to sections 4(i), 7(a), 303, 308(b), and 316 of the Communications Act of 1934, as amended, 47 U.S.C. §§ 154(i), 157(a), 303, 308(b), 316.

C. Description and Estimate of the Number of Small Entities to Which the Proposed Rules will Apply

5. The RFA directs agencies to provide a description of, and where feasible, an estimate of the number of small entities that may be affected by the proposed rules and policies, if adopted.⁷ The RFA generally defines the term “small entity” as having the same meaning as the terms “small business,” “small organization,” and “small governmental jurisdiction.”⁸ In addition, the term “small business” has the same meaning as the term “small business concern” under the Small Business Act.⁹ A “small business concern” is one which: (1) is independently owned and operated; (2) is not dominant in its field of operation; and (3) satisfies any additional criteria established by the SBA.¹⁰

6. *Satellite Telecommunications.* This industry comprises firms “primarily engaged in providing telecommunications services to other establishments in the telecommunications and broadcasting industries by forwarding and receiving communications signals via a system of satellites or reselling satellite telecommunications.”¹¹ Satellite telecommunications service providers include satellite and earth station operators. The SBA small business size standard for this industry classifies a business with \$44 million or less in annual receipts as small.¹² U.S. Census Bureau data for 2017 show that 275 firms in this industry operated for the entire year.¹³ Of this number, 242 firms had revenue of less than \$25 million.¹⁴ Consequently, using the SBA’s small business size standard most satellite telecommunications service providers can be considered small entities. The Commission notes however, that the SBA’s revenue small business size standard is applicable to a broad scope of satellite telecommunications providers included in the U.S. Census Bureau’s Satellite Telecommunications industry definition. Additionally, the Commission neither requests nor collects annual revenue information from satellite telecommunications providers, and is therefore unable to more accurately estimate the number of satellite telecommunications providers that would be classified as a small business under the SBA size standard.

7. *All Other Telecommunications.* This industry is comprised of establishments primarily

⁷ 5 U.S.C. § 603(b)(3).

⁸ *Id.* § 601(6).

⁹ *Id.* § 601(3) (incorporating by reference the definition of “small-business concern” in the Small Business Act, 15 U.S.C. § 632). Pursuant to 5 U.S.C. § 601(3), the statutory definition of a small business applies “unless an agency, after consultation with the Office of Advocacy of the Small Business Administration and after opportunity for public comment, establishes one or more definitions of such term which are appropriate to the activities of the agency and publishes such definition(s) in the Federal Register.”

¹⁰ 15 U.S.C. § 632.

¹¹ See U.S. Census Bureau, *2017 NAICS Definition, “517410 Satellite Telecommunications,”* <https://www.census.gov/naics/?input=517410&year=2017&details=517410>.

¹² See 13 CFR § 121.201, NAICS Code 517410.

¹³ See U.S. Census Bureau, *2017 Economic Census of the United States, Selected Sectors: Sales, Value of Shipments, or Revenue Size of Firms for the U.S.: 2017*, Table ID: EC1700SIZEREVFIRM, NAICS Code 517410, <https://data.census.gov/cedsci/table?y=2017&n=517410&tid=ECNSIZE2017.EC1700SIZEREVFIRM&hidePreview=false>. At this time, the 2022 Economic Census data is not available.

¹⁴ *Id.* The available U.S. Census Bureau data does not provide a more precise estimate of the number of firms that meet the SBA size standard. We also note that according to the U.S. Census Bureau glossary, the terms receipts and revenues are used interchangeably, see https://www.census.gov/glossary/#term_ReceiptsRevenueServices.

engaged in providing specialized telecommunications services, such as satellite tracking, communications telemetry, and radar station operation.¹⁵ This industry also includes establishments primarily engaged in providing satellite terminal stations and associated facilities connected with one or more terrestrial systems and capable of transmitting telecommunications to, and receiving telecommunications from, satellite systems.¹⁶ Providers of Internet services (e.g. dial-up ISPs) or Voice over Internet Protocol (VoIP) services, via client-supplied telecommunications connections are also included in this industry.¹⁷ The SBA small business size standard for this industry classifies firms with annual receipts of \$40 million or less as small.¹⁸ U.S. Census Bureau data for 2017 show that there were 1,079 firms in this industry that operated for the entire year.¹⁹ Of those firms, 1,039 had revenue of less than \$25 million.²⁰ Based on this data, the Commission estimates that the majority of “All Other Telecommunications” firms can be considered small.

D. Description of Economic Impact and Projected Reporting, Recordkeeping, and Other Compliance Requirements for Small Entities

8. The RFA directs agencies to provide a description of the projected reporting, recordkeeping and other compliance requirements of the proposed rule, including an estimate of the classes of small entities which will be subject to the requirement and the type of professional skills necessary for preparation of the report or record.²¹

9. The *NPRM* seeks to develop a robust technical record on the current NGSO-GSO spectrum sharing regime in the 10.7-12.7, 17.3-18.6, and 19.7-20.2 GHz bands, and how it might be updated or improved in light of modern satellite technologies and spectrum management techniques. The *NPRM* does not propose any reporting, recordkeeping, or other compliance requirements. However, matters upon which the *NPRM* seeks comment could result in new and/or modified reporting, recordkeeping, or other compliance requirements for small and other NGSO and GSO operators. For example, the *NPRM* seeks comment on what, if any, information GSO operators should be required to share to facilitate the efficient use of spectrum, and whether NGSO operators should be required to demonstrate compliance with any new NGSO-GSO sharing framework that may be adopted in their initial application or whether authorization should be conditional on later showings of compliance.

10. With regard to the compliance costs for small entities, the *NPRM* initiates a review to take a “fresh-look” at current technology and operations in the satellite industry, thus, at this time the record does not include sufficient cost and/or economic analyses to allow the Commission to quantify the costs of compliance for small entities, including whether it will be necessary for small entities to hire professionals to comply with any rules that may be adopted in this proceeding. Information the Commission receives in comments may include analyses of the costs and benefits of various alternative

¹⁵ See U.S. Census Bureau, *2017 NAICS Definition*, “517919 All Other Telecommunications,” <https://www.census.gov/naics/?input=517919&year=2017&details=517919>.

¹⁶ *Id.*

¹⁷ *Id.*

¹⁸ See 13 CFR § 121.201, NAICS Code 517919 (as of 10/1/22, NAICS Code 517810).

¹⁹ See U.S. Census Bureau, *2017 Economic Census of the United States, Selected Sectors: Sales, Value of Shipments, or Revenue Size of Firms for the U.S.: 2017*, Table ID: EC1700SIZEREVFIRM, NAICS Code 517919, <https://data.census.gov/cedsci/table?y=2017&n=517919&tid=ECNSIZE2017.EC1700SIZEREVFIRM&hidePreview=false>.

²⁰ *Id.* The available U.S. Census Bureau data does not provide a more precise estimate of the number of firms that meet the SBA size standard. We also note that according to the U.S. Census Bureau glossary, the terms receipts and revenues are used interchangeably, see https://www.census.gov/glossary/#term_ReceiptsRevenueServices.

²¹ 5 U.S.C. § 603(b)(4).

NGSO-GSO sharing frameworks and their associated requirements which should help the Commission assess any compliance costs for small entities. Industry input should also allow the Commission to identify and evaluate additional matters and burdens relevant to small entities that may result from the inquiries we make in this proceeding.

E. Discussion of Significant Alternatives That Minimize the Significant Economic Impact on Small Entities

11. The RFA directs agencies to provide a description of any significant alternatives to the proposed rules that would accomplish the stated objectives of applicable statutes, and minimize any significant economic impact on small entities.²² The discussion is required to include alternatives such as: “(1) the establishment of differing compliance or reporting requirements or timetables that take into account the resources available to small entities; (2) the clarification, consolidation, or simplification of compliance and reporting requirements under the rules for such small entities; (3) the use of performance rather than design standards; and (4) an exemption from coverage of the rule, or any part thereof, for such small entities.”²³

12. The *NPRM* seeks comment on various alternatives to develop the most efficient and effective use of the shared spectrum in the 10.7-12.7, 17.3-18.6, and 19.7-20.2 GHz bands, and to assess any alternative NGSO-GSO sharing framework in the these bands proposed by commenters on the *NPRM*. For example, the *NPRM* seeking input on the appropriate methodology for protecting GSO operations that use Adaptive Coding and Modulation (ACM) inquires whether NGSO operators should be afforded the same flexibility to adjust their operations to meet any new methodology and thresholds similar to what the Commission provided in the NGSO-NGSO sharing context. The *NPRM* also seeks to explore the appropriate methodology for protecting GSO operations that do not use ACM and requests comment on whether short-term protection criterion should be the only protection criterion, or alternatively, whether a long-term protection criterion such as an interference-to-noise (I/N) threshold may be a more appropriate threshold for GSO links that do not use ACM. Regarding what modeling assumptions should underlie any methodology, the *NPRM* raises alternatives such as whether the Commission should adopt a minimum link availability threshold to guard against risk of low-performing links; adopt standardized reference carrier-to-noise (C/N) threshold values to account for the performance of efficient, modern modems and receivers; adopt a minimum receiver performance standard, such as a limit on the receiver noise temperature; incorporate interference from other GSO operations into the baseline calculations for any methodology; or require the use of particular rain-fade characteristics, such as those found in Recommendations ITU-R P.618 or P.676.

13. The alternative NGSO-GSO spectrum sharing framework inquiry in the *NPRM* requests input on whether the Commission should continue to require NGSO operators not to cause unacceptable interference into GSO networks, absent coordination, or in the alternative require GSO operators to not cause unacceptable interference into NGSO systems, absent coordination. The *NPRM* also seeks comment on whether any protections offered to GSO operations should be uniform in scope, or should they be tailored to the actual usage in particular orbital slots. As part of the record the Commission seeks to develop on information sharing, and what if any, information GSO operators should be required to share to facilitate the efficient use of spectrum, the *NPRM* also inquires whether GSO operators should be required to share certain information including the actual antenna patterns of their deployed earth stations to the extent they differ from those in public records; their actual power levels or other system characteristics that could impact their baseline operations; the actual spectrum used or spectrum usage patterns of particular satellites to maximize spectral efficiency; the identity of links that require particular interference protection as well as the basis for any such heightened need; and whether their networks rely

²² *Id.* § 603(c).

²³ 5 U.S.C. § 603(c)(1)-(4).

on ACM and, if not, the link budgets they use.

14. Finally, the *NPRM*'s request for comment involving compliance with, and the transition to, any new NGSO-GSO sharing framework seeks feedback on whether NGSO operators should be required to demonstrate compliance in their initial application, or instead should authorization be conditional on later showings of compliance; whether the Commission should sunset any protections for inefficient or outdated technologies or services; and whether to offer current licensees and market access grantees a simple process to request modification of their authorizations to conform with any new rules adopted, such as allowing requests to be made by letter. All of the above alternatives upon which comment is sought could result in requirements that have an economic impact on small entities.

15. The Commission expects to fully consider any proposals, and comments, including costs analyses submitted on the record in response to the *NPRM*, and assess whether they would have a significant economic impact on small entities. Information from industry stakeholders to create a robust technical record and provide responses to the inquiries in the *NPRM* will allow the Commission to better evaluate options and alternatives for minimization of any significant economic impact on small entities. The Commission's evaluation of this information will shape the final alternatives it considers to minimize any significant economic impact that may occur on small entities, the final conclusions it reaches, and any final rules it promulgates in this proceeding.

F. Federal Rules that May Duplicate, Overlap, or Conflict with the Proposed Rules

16. None.

APPENDIX B
List of Commenters

Comments and Oppositions in RM-11990

CTIA – The Wireless Association (CTIA)
EchoStar Corporation (EchoStar)
Eutelsat S.A. and WorldVu Satellites Limited (collectively, Eutelsat)
Kuiper Systems LLC (Kuiper)
SES Americom, Inc. and O3b Limited (collectively, SES)
Space Exploration Holdings, LLC (SpaceX)
Telesat Canada (Telesat)
JSX
Viasat, Inc. (Viasat)

Reply Comments in RM-11990

Cadets in the National Space Policy class at the United States Air Force Academy (USFA Cadets)
DIRECTV, LLC (DIRECTV)
EchoStar
Kuiper
SES
SpaceX
Telesat
ViaSat

Ex parte filings in RM-11990

CTIA November 5, 2024 *ex parte*
Eutelsat November 5, 2024 *ex parte*
SpaceX December 10, 2024 *ex parte*
Kuiper January 31, 2025 *ex parte*